NEET UG PHYSICS TUTORIAL

Oscillations and Simple Harmonic Motion

Complete Conceptual Guide, Mathematical Formulations, Worked Examples, and NEET-Aligned Practice Assessment

1. Learning Objectives & Chapter Overview

In everyday life, we encounter various kinds of motions. Rectilinear motion and projectile motion are non-repetitive. In contrast, uniform circular motion and the orbital motion of planets in the solar system repeat after regular time intervals and are called **periodic motions**. When an object moves back and forth (to and fro) repeatedly about a fixed mean position—such as a rocking cradle, a swinging pendulum of a wall clock, a boat tossing up and down on a river, or a reciprocating piston in a steam engine—the motion is termed **oscillatory motion**.

Core Learning Objectives for NEET UG:

- Distinguish clearly between periodic motion, oscillatory motion, and simple harmonic motion (SHM).
- Master fundamental parameters: time period (T), frequency (ν), angular frequency (ω), amplitude (A), phase ($\omega t + \phi$), and phase constant (ϕ).
- Analyze periodic functions using Fourier superposition principles and trigonometric combinations.
- Visualize SHM as the projection of uniform circular motion on a diameter of the reference circle.
- Derive and apply instantaneous and extreme relations for displacement $x(t)$, velocity $v(t)$, and acceleration $a(t)$.
- Formulate the restoring force law $F = -kx$ and calculate kinetic energy K, potential energy U, and total mechanical energy E in SHM.
- Determine the time period of spring-mass combinations, simple pendulums, floating cylinders, and liquid columns in U-tubes.

Oscillatory motion is foundational to physics: vibrating strings in musical instruments (sitar, guitar, violin), vibrating membranes in drums and speaker diaphragms, air molecule vibrations propagating sound waves, atomic vibrations in solids (whose average vibrational energy is proportional to temperature), and alternating current (AC) voltage oscillating positive and negative about zero all rely on these principles. Furthermore, any material medium can be pictured as a collection of coupled oscillators whose collective oscillations manifest as waves.

2. Periodic and Oscillatory Motions

A motion that repeats itself at regular intervals of time is called **periodic motion**. Consider three illustrative physical situations:

1. An insect climbs up a ramp at a steady rate, falls down vertically to the initial point, and repeats the process identically.
2. A child climbs up a step, stays on it briefly, comes down, waits briefly, and repeats the cycle identically.
3. A ball is bounced repeatedly between a player's palm and the ground. Both the downward and upward curved sections of the height-versus-time graph are sections of a parabola governed by Newton's kinematic equations:

$$h = ut + \frac{1}{2}gt^2 \quad (\text{for downward motion})$$

$$h = ut - \frac{1}{2}gt^2 \quad (\text{for upward motion})$$

with appropriate values of initial velocity u in each phase.



Tutorial Diagram 1: Examples of periodic motion showing repeating displacement profiles with period T.

Equilibrium Position, Oscillations, and Vibrations

Very often, a body undergoing periodic motion has an **equilibrium position** (mean position) inside its path where no net external force acts on it ($\mathbf{F}_{\text{net}} = 0$). If placed at rest at this point, it remains there forever. When given a small displacement away from this position, a **restoring force** arises that tries to bring the body back toward the equilibrium point, producing **oscillations** or **vibrations** (for example, a ball oscillating at the bottom of a smooth curved bowl).

- **Crucial Distinction:** Every oscillatory motion is periodic, but *every periodic motion need not be oscillatory*. For instance, uniform circular motion or planetary orbital motion is periodic, but it is not oscillatory because it does not move to and fro about a mean equilibrium point.
- **Oscillation vs. Vibration:** There is no fundamental physical difference between the two terms. When the frequency is small (such as the swaying branch of a tree), we call it an *oscillation*; when the frequency is high (such as the string of a musical instrument), we call it a *vibration*.
- **Damped and Forced Oscillations:** In real life, oscillating bodies eventually come to rest at their equilibrium positions due to damping caused by friction and other dissipative forces. However, they can be maintained in oscillation by applying an external periodic agency (forced oscillations).

2.1 Period and Frequency

- **Period (T):** The smallest interval of time after which a periodic motion repeats itself is called its **period**. Its SI unit is the second (s). For very fast or very slow motions, other convenient units are used: the vibration period of a quartz crystal is measured in microseconds ($1\,\mu\mathrm{s} = 10^{-6}\,\mathrm{s}$), the orbital period of planet Mercury is $88\ \mathrm{Earth\ days}$, and Halley's comet reappears every $76\ \mathrm{years}$.
- **Frequency (ν):** The number of repetitions (oscillations or cycles) occurring per unit time is called the **frequency** of the periodic motion. It is the reciprocal of the time period:

$$\nu = \frac{1}{T}$$

The SI unit of frequency is s^{-1}, given the special name **hertz (Hz)** in honour of Heinrich Rudolph Hertz (1857–1894), the discoverer of radio waves:

$$1\,\mathrm{hertz} = 1\,\mathrm{Hz} = 1\ \mathrm{oscillation\ per\ second} = 1\,\mathrm{s}^{-1}$$

Note that ν is a positive real quantity and is *not* necessarily an integer.

Worked Example 1: Frequency and Time Period of the Human Heart

Problem: On an average, a human heart is found to beat $75\ \text{times}$ in a minute. Calculate its beat frequency and period.

Solution:

Number of beats in $1\ \text{minute}\ (60\,\text{s}) = 75$.

The beat frequency ν is the number of repetitions per second:

$$\nu = \frac{75}{1\,\text{min}} = \frac{75}{60\,\text{s}} = 1.25\,\text{s}^{-1} = 1.25\,\text{Hz}$$

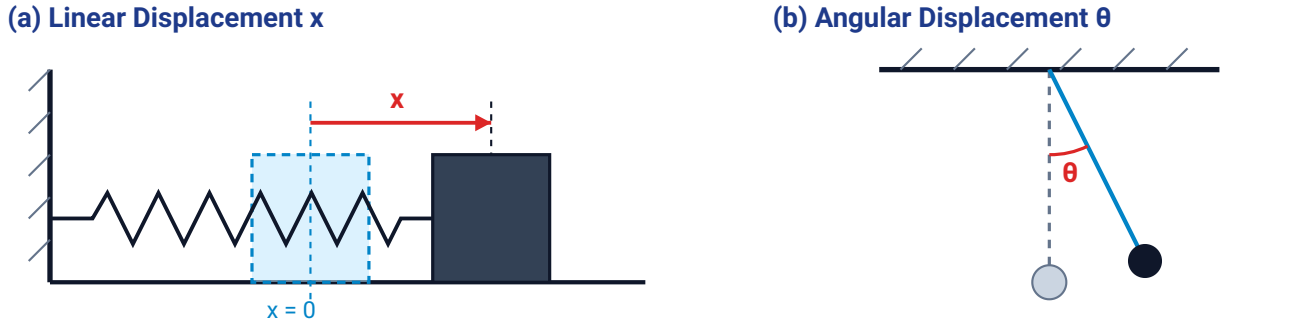
The time period T is the reciprocal of frequency:

$$T = \frac{1}{\nu} = \frac{1}{1.25\,\text{s}^{-1}} = 0.8\,\text{s}$$

2.2 Generalized Concept of Displacement and Periodic Functions

In oscillatory motion, the term **displacement** is used in a generalized sense to denote the change with time of any physical property under consideration, usually measured from its convenient equilibrium value:

- For a block attached to a spring on a frictionless surface (with the other end fixed to a rigid wall), the linear distance $x(t)$ from the unstretched equilibrium position is the displacement variable.
- For an oscillating simple pendulum, the angular deflection $\theta(t)$ from the vertical is the displacement variable.
- In an AC circuit, the time-varying voltage across a capacitor is a displacement variable.
- In sound propagation, local pressure variations in time act as the displacement variable; in light waves, the time-varying electric and magnetic fields serve as displacement variables.



Tutorial Diagram 2: Displacement variables in (a) a horizontal spring-block oscillator and (b) a simple pendulum.

Mathematically, a periodic motion is represented by a periodic function $f(t)$ satisfying $f(t) = f(t+T)$. The simplest periodic functions are sine and cosine functions:

$$f(t) = A\cos\omega t \quad \text{or} \quad f(t) = A\sin\omega t$$

Since $\cos(\theta + 2\pi) = \cos\theta$, increasing the argument ωt by $2\pi\ \text{radians}$ leaves the function unchanged. Thus, the period T satisfies $\omega(t+T) = \omega t + 2\pi$, yielding:

$$T = \frac{2\pi}{\omega} \quad \Longleftrightarrow \quad \omega = \frac{2\pi}{T} = 2\pi\nu$$

Furthermore, any linear combination of sine and cosine functions of the same angular frequency ω is also periodic with the same period T:

$$f(t) = A\sin\omega t + B\cos\omega t$$

By substituting $A = D\cos\phi$ and $B = D\sin\phi$, this linear combination simplifies into a single sinusoidal wave:

$$f(t) = D\sin(\omega t + \phi)$$

$$\text{where resultant amplitude } D = \sqrt{A^2 + B^2} \quad \text{and initial phase } \phi = \tan^{-1}\left(\frac{B}{A}\right)$$

Fourier's Theorem: The French mathematician Jean Baptiste Joseph Fourier (1768–1830) proved a remarkable theorem: *Any periodic function can be expressed as a superposition of sine and cosine functions of different time periods (integral multiples of the fundamental frequency) with suitable coefficients.*

Worked Example 2: Identifying Periodic and Non-Periodic Functions

Problem: Which of the following functions of time represent (a) periodic and (b) non-periodic motion? Give the period for each case of periodic motion (ω is any positive constant):

i. $\sin\omega t + \cos\omega t$
ii. $\sin\omega t + \cos 2\omega t + \sin 4\omega t$
iii. $e^{-\omega t}$
iv. $\log(\omega t)$

Solution:

(i) $\sin\omega t + \cos\omega t = \sqrt{1^2+1^2}\,\sin\left(\omega t + \frac{\pi}{4}\right) = \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$. Since $\sqrt{2}\sin\left(\omega t + \frac{\pi}{4} + 2\pi\right) = \sqrt{2}\sin\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \frac{\pi}{4}\right]$, this is a **periodic function** (and specifically SHM) with time period $T = \frac{2\pi}{\omega}$.

(ii) Each individual term is periodic with its own period:
• $\sin\omega t$ has period $T_0 = \frac{2\pi}{\omega}$.
• $\cos 2\omega t$ has period $T_2 = \frac{2\pi}{2\omega} = \frac{\pi}{\omega} = \frac{T_0}{2}$.
• $\sin 4\omega t$ has period $T_3 = \frac{2\pi}{4\omega} = \frac{T_0}{4}$.
Since the period of the first term (T_0) is an integral multiple of the periods of the other two terms ($T_0 = 2T_2 = 4T_3$), the smallest interval of time after which the entire sum repeats is T_0. Thus, the sum represents **periodic motion** with period $T = \frac{2\pi}{\omega}$.

(iii) The function $e^{-\omega t}$ decreases monotonically with increasing time t and approaches 0 as $t \to \infty$. It never repeats its value, so it represents **non-periodic motion**.

(iv) The function $\log(\omega t)$ increases monotonically with time t and diverges to ∞ as $t \to \infty$. It never repeats its value and is **non-periodic** (and cannot represent any physical bounded displacement).

3. Simple Harmonic Motion (SHM)

Consider a particle oscillating back and forth about the origin $x = 0$ along the x-axis between the extreme limits $+A$ and $-A$. This oscillatory motion is defined as **Simple Harmonic Motion (SHM)** if the displacement x of the particle from the origin varies sinusoidally with time:

$$x(t) = A\cos(\omega t + \phi)$$

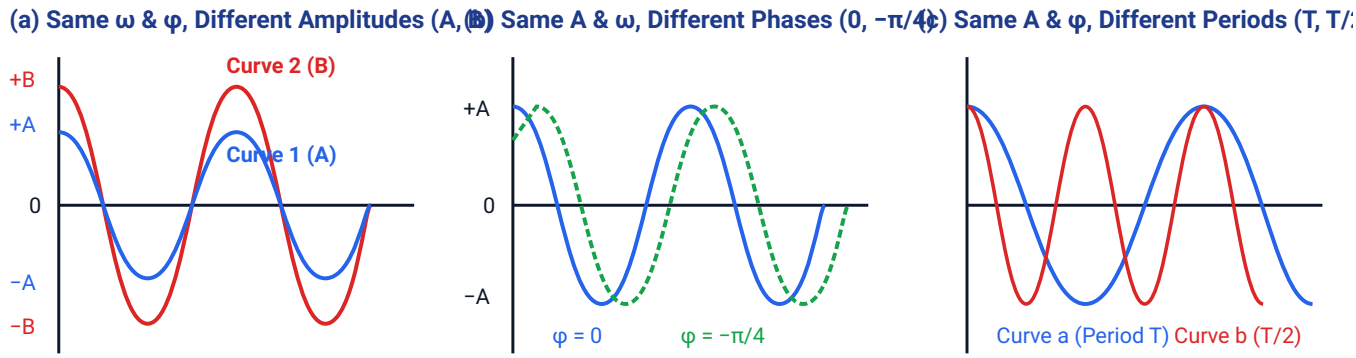
Here A, ω, and ϕ are constants characterizing the given SHM:

Symbol	Standard Name	Physical Meaning & Properties
$x(t)$	Instantaneous Displacement	Position of the oscillating particle from the mean position ($x = 0$) at instant t. Ranges between $-A$ and $+A$.
A	Amplitude	Magnitude of the maximum displacement of the particle on either side of the mean position ($A > 0$).
ω	Angular Frequency	Rate of change of phase with time: $\omega = \frac{2\pi}{T} = 2\pi\nu$. SI unit: $\mathrm{rad\,s^{-1}}$.
$(\omega t + \phi)$	Phase	Time-dependent argument of the cosine/sine function that uniquely determines the state of motion (both position and velocity) at any instant t.
ϕ	Phase Constant (Initial Phase / Epoch)	Value of the phase at $t = 0$. Determined by initial displacement $x(0)$ and initial velocity $v(0)$.

Discrete Snapshots and Continuous Graphs of SHM

For a particle starting at $x = +A$ at $t = 0$ (so $\phi = 0$ and $x(t) = A\cos\omega t$), its position at discrete quarter-period intervals $T/4$ illustrates how the motion unfolds:

- At $t = 0$: Particle is at the positive extreme $x = +A$, instantaneous speed $v = 0$.
- At $t = T/4$: Particle passes through the mean position $x = 0$ moving left with maximum speed $v_{\max}$.
- At $t = T/2$: Particle reaches the negative extreme $x = -A$, instantaneous speed $v = 0$.
- At $t = 3T/4$: Particle passes through the mean position $x = 0$ moving right with maximum speed $v_{\max}$.
- At $t = T$: Particle returns to $x = +A$, completing one full oscillation cycle.



Tutorial Diagram 3: Comparison of SHM displacement-time graphs varying (a) amplitude, (b) phase constant, and (c) time period.

Worked Example 3: Distinguishing SHM from Shifted Harmonic/Periodic Motion

Problem: Which of the following functions of time represent (a) simple harmonic motion about $x = 0$ and (b) periodic motion with a shifted equilibrium point? Give the period for each case:

1. $\sin \omega t - \cos \omega t$
2. $\sin^2 \omega t$

Solution:

(1) $\sin \omega t - \cos \omega t = \sin \omega t - \sin\left(\frac{\pi}{2} - \omega t\right) = 2\cos\left(\frac{\pi}{4}\right)\sin\left(\omega t - \frac{\pi}{4}\right) = \sqrt{2}\sin\left(\omega t - \frac{\pi}{4}\right)$.
This is a single sinusoidal function oscillating symmetrically about 0. Hence, it represents **Simple Harmonic Motion** with amplitude $A = \sqrt{2}$, period $T = \frac{2\pi}{\omega}$, and phase angle $-\frac{\pi}{4}$ (or equivalently $\frac{7\pi}{4}$).

(2) Using the half-angle identity $\sin^2 \theta = \frac{1-\cos 2\theta}{2}$, we write:

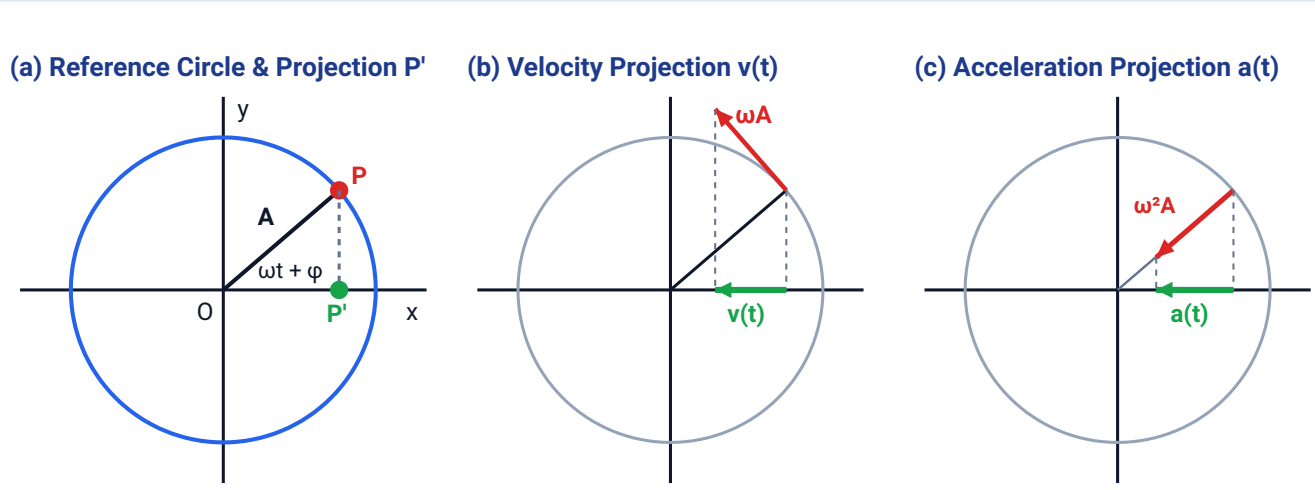
$$\sin^2 \omega t = \frac{1}{2} - \frac{1}{2}\cos(2\omega t)$$

The angular frequency of the oscillating term is 2ω, so the function is **periodic** with period $T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$. Because of the constant offset $\frac{1}{2}$, it does not oscillate about $x = 0$; rather, it represents harmonic motion with the equilibrium position shifted to $\frac{1}{2}$ instead of zero (oscillating between 0 and 1).

4. Simple Harmonic Motion and Uniform Circular Motion

Theorem: *The projection of uniform circular motion on any diameter of the circle executes Simple Harmonic Motion.*

Imagine tying a ball to the end of a string and making it revolve in a horizontal plane at a constant angular speed ω. If you view the ball edge-on (sideways in the plane of the circle) or observe its shadow cast onto a vertical wall by a distant parallel light beam, the ball appears to move back and forth along a straight diameter. The revolving particle P is called the **reference particle**, and the circle of radius A on which it moves is called the **reference circle**.



Tutorial Diagram 4: Geometric derivation of (a) position x(t), (b) velocity v(t), and (c) acceleration a(t) from the reference circle.

Suppose the reference particle P moves uniformly anticlockwise on a circle of radius A with constant angular speed ω. At $t = 0$, the position vector $\mathbf{OP}$ makes an initial angle ϕ with the positive x-axis. In time t, it sweeps through an additional angle ωt, so that $\mathbf{OP}$ makes an angle $(\omega t + \phi)$ with the $+x$-axis:

- **Projection on the x-axis (P'):**

$$x(t) = A\cos(\omega t + \phi)$$

- **Projection on the y-axis:**

$$y(t) = A\sin(\omega t + \phi) = A\cos\left(\omega t + \phi - \frac{\pi}{2}\right)$$

Thus, the y-projection also executes SHM with the same amplitude A and angular frequency ω, but lags behind the x-projection by a phase angle of $\frac{\pi}{2}$.

Important Note on Units of Angle: The natural unit of angle is the radian (rad), defined as the ratio of arc length to radius (dimensionless). Whenever the argument of a trigonometric function is written without explicit units, it is understood to be in radians. For instance, $\sin(15^\circ)$ means the sine of 15 degrees, whereas $\sin(15)$ means the sine of 15 radians.

Worked Example 4: Writing SHM Equations from Reference Circle Diagrams

Problem: Consider two uniform circular motions of a reference particle P:

- **Case (a):** Radius of circle $= A$, period of revolution $T = 4\,\mathrm{s}$, sense of rotation is **anticlockwise**, and at $t = 0$ the radius vector $\mathbf{OP}$ makes an angle of 45° ($\pi/4\,\mathrm{rad}$) in the first quadrant with the $+x$-axis.
- **Case (b):** Radius of circle $= B$, period of revolution $T = 30\,\mathrm{s}$, sense of rotation is **clockwise**, and at $t = 0$ the particle P is at the topmost point on the $+y$-axis (90° from the $+x$-axis).

Obtain the simple harmonic motion equation of the x-projection of the radius vector of the rotating particle P in each case.

Solution:

(a) At $t = 0$, $\mathbf{OP}$ makes an angle $\phi = 45^\circ = \frac{\pi}{4}\,\mathrm{rad}$ with the $+x$-axis. In time t, rotating anticlockwise, it covers an angle $\frac{2\pi}{T}t$ and makes a total angle $\left(\frac{2\pi}{T}t + \frac{\pi}{4}\right)$ with the $+x$-axis. Substituting $T = 4\,\mathrm{s}$:

$$x(t) = A\cos\left(\frac{2\pi}{4}t + \frac{\pi}{4}\right) = A\cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$$

This represents SHM of amplitude A, period $4\,\mathrm{s}$, and initial phase $\phi = +\frac{\pi}{4}$.

(b) At $t = 0$, $\mathbf{OP}$ makes an angle of $90^\circ = \frac{\pi}{2}\,\mathrm{rad}$ with the $+x$-axis. Rotating *clockwise*, after time t the angle with the $+x$-axis decreases by $\frac{2\pi}{T}t$, becoming $\left(\frac{\pi}{2} - \frac{2\pi}{T}t\right)$. Projecting onto the x-axis for $T = 30\,\mathrm{s}$:

$$x(t) = B\cos\left(\frac{\pi}{2} - \frac{2\pi}{30}t\right) = B\sin\left(\frac{\pi}{15}t\right) = B\cos\left(\frac{\pi}{15}t - \frac{\pi}{2}\right)$$

This represents SHM of amplitude B, period $30\,\mathrm{s}$, and initial phase $\phi = -\frac{\pi}{2}$.

5. Velocity and Acceleration in Simple Harmonic Motion

We can obtain the instantaneous velocity and acceleration of a particle in SHM both geometrically (from the reference circle) and analytically (by calculus differentiation).

5.1 Instantaneous Velocity $v(t)$

In uniform circular motion, the tangential speed of the reference particle P is $v = \omega A$. At angle $(\omega t + \phi)$ in the first quadrant, the tangential velocity vector points upward and to the left, so its projection along the x-axis is directed in the negative x-direction. Equivalently, differentiating $x(t) = A\cos(\omega t + \phi)$ with respect to time t:

$$v(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} = -\omega A\sin(\omega t + \phi)$$

Using the trigonometric identity $\sin^2(\omega t+\phi) = 1-\cos^2(\omega t+\phi) = 1-\frac{x^2}{A^2}$, we get velocity as a function of position x:

$$v(x) = \pm\omega\sqrt{A^2-x^2}$$

Velocity Amplitude (Maximum Speed): $v_{\max} = \omega A$ (occurs at the mean position $x=0$). At the extreme positions $x=\pm A$, velocity is zero ($v=0$).

5.2 Instantaneous Acceleration $a(t)$

The reference particle P in uniform circular motion experiences a radial centripetal acceleration of magnitude $\frac{v^2}{A} = \omega^2 A$ directed radially inward along $\mathbf{PO}$. Its projection on the x-axis (or direct differentiation of $v(t)$ with respect to time t) gives:

$$a(t) = \frac{\mathrm{d}v(t)}{\mathrm{d}t} = -\omega^2 A\cos(\omega t+\phi) = -\omega^2 x(t)$$

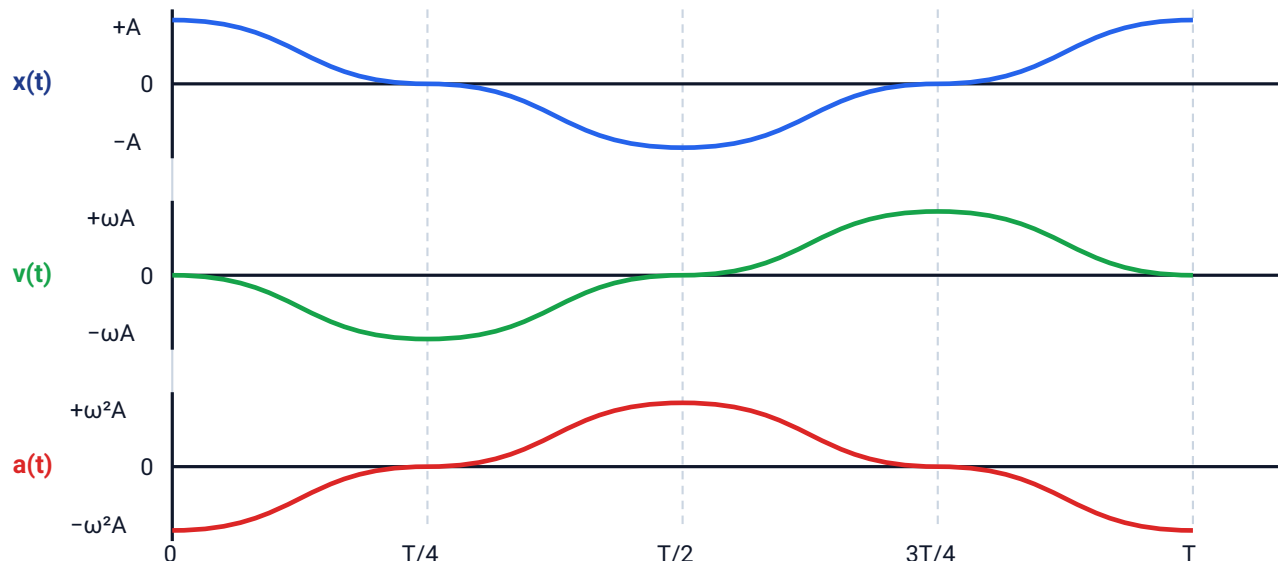
Acceleration Amplitude (Maximum Acceleration): $a_{\max} = \omega^2 A$ (occurs at the extreme positions $x=\pm A$). At the mean position $x=0$, acceleration is zero ($a=0$).

Crucial Property of SHM: In SHM, instantaneous acceleration is directly proportional to displacement $x(t)$ and is always directed opposite to displacement (towards the mean position $x=0$): when $x(t)>0$, $a(t)<0$; when $x(t)<0$, $a(t)>0$.

5.3 Phase Relationships Between Displacement, Velocity, and Acceleration

Setting $\phi = 0$ for simplicity:

- $x(t) = A\cos\omega t$
- $v(t) = -\omega A\sin\omega t = \omega A\cos\left(\omega t+\frac{\pi}{2}\right) \implies$ **Velocity leads displacement by a phase angle of $\frac{\pi}{2}$ (90°).**
- $a(t) = -\omega^2 A\cos\omega t = \omega^2 A\cos(\omega t+\pi) \implies$ **Acceleration leads displacement by a phase angle of π (180°)** (they are in opposite phase, or antiphase), and acceleration leads velocity by $\frac{\pi}{2}$.



Tutorial Diagram 5: Simultaneous time plots of displacement x(t), velocity v(t), and acceleration a(t) over one period T.

Worked Example 5: Calculating Displacement, Velocity, and Acceleration at a Given Instant

Problem: A body oscillates with SHM according to the equation (in SI units):

$$x = 5\cos\left(2\pi t + \frac{\pi}{4}\right)$$

At $t = 1.5\,\text{s}$, calculate the (a) displacement, (b) speed, and (c) acceleration of the body.

Solution:

Comparing with $x(t) = A\cos(\omega t + \phi)$, we identify amplitude $A = 5.0\,\text{m}$, angular frequency $\omega = 2\pi\,\text{s}^{-1}$ (so period $T = 1\,\text{s}$), and phase constant $\phi = \frac{\pi}{4}\,\text{rad}$.

At $t = 1.5\,\text{s}$, the phase is:

$$\omega t + \phi = (2\pi\,\text{s}^{-1})(1.5\,\text{s}) + \frac{\pi}{4} = 3\pi + \frac{\pi}{4}$$

Since $\cos(3\pi + \theta) = -\cos\theta$ and $\sin(3\pi + \theta) = -\sin\theta$:

(a) Displacement:

$$x(1.5\,\text{s}) = (5.0\,\text{m})\cos\left(3\pi + \frac{\pi}{4}\right) = -5.0\cos\left(\frac{\pi}{4}\right) = -5.0 \times 0.707\,\text{m} = -3.535\,\text{m}$$

(b) Velocity and Speed:

$$v(1.5\,\text{s}) = -\omega A\sin\left(3\pi + \frac{\pi}{4}\right) = -(2\pi\,\text{s}^{-1})(5.0\,\text{m})\left(-\sin\frac{\pi}{4}\right) = 10\pi \times 0.707\,\text{m}\,\text{s}^{-1} \approx 22\,\text{m}\,\text{s}^{-1}$$

(c) Acceleration:

$$a(1.5\,\text{s}) = -\omega^2 x(1.5\,\text{s}) = -(2\pi\,\text{s}^{-1})^2 \times (-3.535\,\text{m}) = 4\pi^2 \times 3.535\,\text{m}\,\text{s}^{-2} \approx 140\,\text{m}\,\text{s}^{-2}$$

6. Force Law for Simple Harmonic Motion

Applying Newton's second law of motion ($F = ma$) together with the SHM acceleration relation $a(t) = -\omega^2 x(t)$ for a particle of mass m:

$$F(t) = ma(t) = -m\omega^2 x(t) = -kx(t)$$

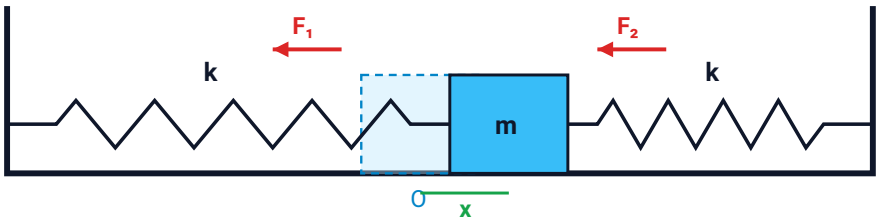
where the force constant (or spring stiffness constant) k is related to mass m and angular frequency ω by:

$$k = m\omega^2 \quad \Longleftrightarrow \quad \omega = \sqrt{\frac{k}{m}} \quad \Longleftrightarrow \quad T = 2\pi\sqrt{\frac{m}{k}}$$

- Because $F(t) = -kx(t)$ is always directed opposite to displacement (towards the equilibrium position $x = 0$), it is called the **restoring force**.
- SHM can be defined in two completely equivalent ways: kinematically by $x(t) = A\cos(\omega t + \phi)$, or dynamically by the linear restoring force law $F(t) = -kx(t)$ (differentiating $x(t)$ twice gives $F(t)$; integrating $F(t)$ twice gives $x(t)$).
- **Linear vs. Non-Linear Oscillators:** An oscillator governed strictly by $F = -kx$ (linearly proportional to x) is called a **linear harmonic oscillator**. In real physical systems, the restoring force may contain small additional higher-order terms proportional to x^2, x^3, etc.; such systems are called **non-linear oscillators**.

Worked Example 6: Block Connected Between Two Identical Springs

Problem: Two identical springs, each of spring constant k, are attached to a block of mass m and to fixed supports on either side on a frictionless horizontal surface. Show that when the mass is displaced from its equilibrium position on either side, it executes simple harmonic motion, and find the period of oscillations.



Solution:

Let the block of mass m be displaced by a small distance x to the right of its equilibrium position O. In this configuration:

- The left spring is *elongated* by length x and pulls the block to the left (towards O) with force $F_1 = -kx$.
- The right spring is *compressed* by the same length x and pushes the block to the left (also towards O) with force $F_2 = -kx$.

The net restoring force F acting on the mass is:

$$F = F_1 + F_2 = -kx - kx = -2kx$$

Since the net force is directly proportional to displacement x and directed towards the mean position, the motion is **Simple Harmonic Motion** with effective force constant $k_{\text{eff}} = 2k$ (parallel combination of springs). Therefore, the time period of oscillation is:

$$T = 2\pi\sqrt{\frac{m}{k_{\text{eff}}}} = 2\pi\sqrt{\frac{m}{2k}}$$

7. Energy in Simple Harmonic Motion

A particle executing SHM possesses both kinetic energy K and potential energy U, both of which vary between zero and their maximum value E.

7.1 Kinetic Energy (K)

Using $v(t) = -\omega A\sin(\omega t + \phi)$ and $m\omega^2 = k$:

$$K(t) = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\sin^2(\omega t + \phi) = \frac{1}{2}kA^2\sin^2(\omega t + \phi)$$

$$K(x) = \frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}k(A^2 - x^2)$$

- K is **maximum** ($\frac{1}{2}kA^2$) at the mean position ($x = 0$) and **zero** at the extreme positions ($x = \pm A$).
- Because $\sin^2(\omega t + \phi) = \frac{1-\cos(2\omega t + 2\phi)}{2}$, kinetic energy oscillates with angular frequency 2ω, frequency 2ν, and **time period** $\frac{T}{2}$.

7.2 Potential Energy (U)

The linear restoring force $F = -kx$ is a conservative force. Choosing potential energy to be zero at the equilibrium position $x = 0$:

$$U(x) = -\int_0^x F\,\mathrm{d}x = \int_0^x kx\,\mathrm{d}x = \frac{1}{2}kx^2$$
$$U(t) = \frac{1}{2}kA^2\cos^2(\omega t + \phi)$$

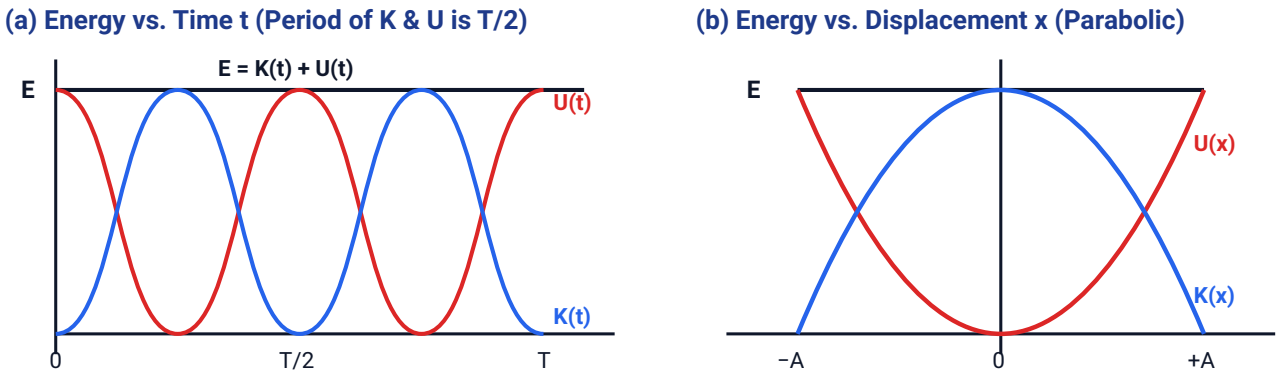
- U is **zero** at the mean position ($x = 0$) and **maximum** ($\frac{1}{2}kA^2$) at the extreme displacements ($x = \pm A$).
- Like kinetic energy, $U(t)$ is always non-negative and repeats with **time period** $\frac{T}{2}$ (frequency 2ν), peaking twice during every complete period T of SHM.

7.3 Total Mechanical Energy (E)

Summing kinetic and potential energies at any time t or position x:

$$E = K + U = \frac{1}{2}kA^2\left[\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)\right] = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2 = 2\pi^2m\nu^2A^2$$

Thus, in the absence of friction/damping, the **total mechanical energy E of a linear harmonic oscillator is conserved**—it is independent of both time t and displacement x.



Tutorial Diagram 6: Variation of kinetic energy K, potential energy U, and total energy E with (a) time t and (b) position x.

Worked Example 7: Kinetic, Potential, and Total Energy of a Spring-Block System

Problem: A block whose mass is $1\,\text{kg}$ is fastened to a spring of spring constant $50\,\text{N}\,\text{m}^{-1}$. The block is pulled to a distance $x = 10\,\text{cm}$ from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$. Calculate the kinetic, potential, and total energies of the block when it is $5\,\text{cm}$ away from the mean position.

Solution:

Given: $m = 1\,\text{kg}$, $k = 50\,\text{N}\,\text{m}^{-1}$, amplitude $A = 10\,\text{cm} = 0.1\,\text{m}$, and instantaneous position $x = 5\,\text{cm} = 0.05\,\text{m}$.

The angular frequency is:

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{50\,\text{N}\,\text{m}^{-1}}{1\,\text{kg}}} = 7.07\,\text{rad}\,\text{s}^{-1}$$

Since the block is released from rest at $x = +0.1\,\text{m}$ at $t = 0$, its displacement is $x(t) = 0.1\cos(7.07t)$. When $x = 0.05\,\text{m}$, $\cos(7.07t) = \frac{0.05}{0.1} = 0.5$, so $\sin(7.07t) = \frac{\sqrt{3}}{2} = 0.866$. The speed of the block at $x = 0.05\,\text{m}$ is:

$$v = \omega A\sin(7.07t) = 7.07 \times 0.1 \times 0.866\,\text{m}\,\text{s}^{-1} = 0.6123\,\text{m}\,\text{s}^{-1}$$

- **Potential Energy (U) at $x = 0.05\,\text{m}$:**

$$U = \frac{1}{2}kx^2 = \frac{1}{2} \times 50\,\text{N}\,\text{m}^{-1} \times (0.05\,\text{m})^2 = 0.0625\,\text{J}$$

- **Kinetic Energy (K) at $x = 0.05\,\text{m}$:**

$$K = \frac{1}{2}mv^2 = \frac{1}{2} \times 1\,\text{kg} \times (0.6123\,\text{m}\,\text{s}^{-1})^2 = 0.1875\,\text{J} \approx 0.19\,\text{J}$$

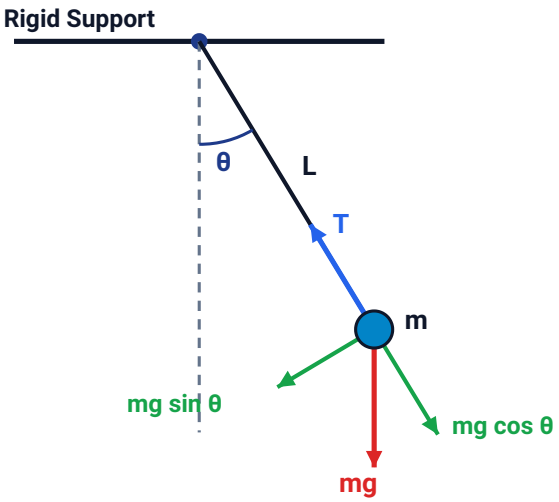
- **Total Mechanical Energy (E):**

$$E = K + U = 0.1875\,\text{J} + 0.0625\,\text{J} = 0.25\,\text{J}$$

(Which matches $E = \frac{1}{2}kA^2 = \frac{1}{2} \times 50 \times (0.1)^2 = 0.25\,\text{J}$, confirming conservation of mechanical energy.)

8. The Simple Pendulum

A **simple pendulum** consists of a small heavy bob of mass m tied to an inextensible, massless string of length L, with the other end fixed to a rigid support so that the bob oscillates freely in a vertical plane about the vertical line through the support.



Tutorial Diagram 7: Free-body diagram of a simple pendulum showing tension T and resolved components of gravity mg.

Derivation of Time Period via Restoring Torque

At angular displacement θ from the vertical mean position, two forces act on the bob: string tension T along the string towards the support, and weight mg vertically downward. Resolving mg:

- The radial component $mg\cos\theta$ acts along the string outward. The net radial force $T - mg\cos\theta$ provides the necessary centripetal acceleration ($\omega_{\text{inst}}^2 L$) for motion along the circular arc, and produces **zero torque** about the support point.
- The tangential component $mg\sin\theta$ acts perpendicular to the string towards the mean position and provides the **restoring torque** τ about the support:

$$\tau = -L(mg\sin\theta)$$

By Newton's rotational law $\tau = I\alpha$, where I is the moment of inertia of the system about the support and α is the angular acceleration:

$$I\alpha = -mgL\sin\theta \quad \Longrightarrow \quad \alpha = -\frac{mgL}{I}\sin\theta$$

Expanding $\sin\theta$ in a power series for θ in radians:

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots$$

Small-Angle Approximation: If the angular displacement θ is small, higher-order powers ($\theta^3, \theta^5, \ldots$) are negligible and $\sin\theta \approx \theta$ (with θ in radians):

θ (degrees)	θ (radians)	$\sin\theta$
0°	0	0
5°	0.087	0.087
10°	0.174	0.174
15°	0.262	0.259
20°	0.349	0.342

Under this small-angle approximation ($\sin\theta \approx \theta$), the angular acceleration equation becomes mathematically identical to the SHM condition $a = -\omega^2 x$:

$$\alpha = -\frac{mgL}{I}\theta = -\omega^2\theta \quad \Longrightarrow \quad \omega = \sqrt{\frac{mgL}{I}} \quad \text{and} \quad T = 2\pi\sqrt{\frac{I}{mgL}}$$

For a simple pendulum with a massless string and point bob, $I = mL^2$, which simplifies to:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Worked Example 8: Length of a Seconds Pendulum

Problem: What is the length of a simple pendulum which ticks seconds (a seconds pendulum)?

Solution:

A simple pendulum that ticks every second takes $1\,\text{s}$ to swing from one extreme to the other, so its complete time period is $T = 2\,\text{s}$. Squaring $T = 2\pi\sqrt{\frac{L}{g}}$ gives:

$$L = \frac{gT^2}{4\pi^2}$$

Substituting $g = 9.8\,\text{m}\,\text{s}^{-2}$ and $T = 2\,\text{s}$:

$$L = \frac{9.8\,\text{m}\,\text{s}^{-2} \times (2\,\text{s})^2}{4\pi^2} = \frac{9.8 \times 4}{4 \times 9.8696}\,\text{m} \approx 1\,\text{m}$$

9. Important Oscillating Systems & Methods for NEET UG

In competitive medical entrance examinations, questions frequently test the application of the restoring force method ($F_{\text{restoring}} = -k_{\text{eff}}x \implies T = 2\pi\sqrt{\frac{m}{k_{\text{eff}}}}$) to diverse physical setups:

9.1 Two-Body Spring Oscillator (Reduced Mass System)

Consider a spring of force constant k attached to a mass m at each free end on a smooth horizontal surface. If each end is pulled outward by the same force F:

- **Maximum Extension:** The tension throughout the massless spring is F (the same as when one end is clamped to a wall and the other is pulled by F). Hence, the maximum extension in *both* cases is simply $x_{\max} = \frac{F}{k}$.
- **Time Period of Oscillation:** When the two equal masses m are released from rest, the centre of the spring remains stationary by symmetry. The spring of length L and constant k behaves as two half-springs in series, each of force constant $2k$, attached to mass m. Thus, each mass experiences $F = -2kx$ and oscillates with period:

  $$T = 2\pi\sqrt{\frac{m}{2k}}$$

  (More generally, for two masses m_1 and m_2 connected by spring k, $T = 2\pi\sqrt{\frac{\mu}{k}}$ where reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$; for $m_1 = m_2 = m$, $\mu = \frac{m}{2}$.)

9.2 Simple Pendulum in an Accelerating Frame (Car on a Circular Track)

When a simple pendulum of length l and bob mass M is suspended inside a car moving with uniform speed v on a horizontal circular track of radius R, the bob experiences two mutually perpendicular pseudo/field accelerations in the frame of the car:

- Downward gravitational acceleration g.
- Horizontal radially outward centrifugal acceleration $a_c = \frac{v^2}{R}$.

Thus, the **effective acceleration due to gravity** g_{eff} is the vector resultant:

$$g_{\text{eff}} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2} \quad \Longrightarrow \quad T = 2\pi\sqrt{\frac{l}{g_{\text{eff}}}} = 2\pi\sqrt{\frac{l}{\sqrt{g^2 + \frac{v^4}{R^2}}}}$$

9.3 Floating Cylinder (Cork) in a Liquid

Consider a cylindrical piece of cork of base area A, height h, and density ρ floating vertically in a liquid of density ρ_l ($\rho < \rho_l$). Mass of the cork is $m = Ah\rho$. If depressed slightly by a vertical downward distance y from equilibrium and released, an additional volume Ay of liquid is displaced, creating an upward excess buoyant restoring force:

$$F_{\text{restoring}} = -(Ay\rho_l)g = -(A\rho_l g)y$$

Here $k_{\text{eff}} = A\rho_l g$. The cork executes SHM with time period:

$$T = 2\pi\sqrt{\frac{m}{k_{\text{eff}}}} = 2\pi\sqrt{\frac{Ah\rho}{A\rho_l g}} = 2\pi\sqrt{\frac{h\rho}{\rho_l g}}$$

9.4 Oscillating Mercury Column in a U-Tube

Consider a U-tube of uniform cross-sectional area A containing a liquid (such as mercury) of density ρ up to height h in each vertical limb (total length of liquid column $L = 2h$, total mass $m = AL\rho = 2Ah\rho$). If a suction pump creates a small level difference and is then removed, when the liquid is depressed by distance y in one limb, it rises by y in the other limb, creating a height difference of $2y$ between the two columns. The excess weight of the column of height $2y$ provides the restoring force:

$$F_{\text{restoring}} = -(A \cdot 2y \cdot \rho)g = -(2A\rho g)y$$

With $k_{\text{eff}} = 2A\rho g$ and $m = 2Ah\rho$, the mercury column executes SHM with time period:

$$T = 2\pi\sqrt{\frac{m}{k_{\text{eff}}}} = 2\pi\sqrt{\frac{2Ah\rho}{2A\rho g}} = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{L}{2g}}$$

Summary Table of Physical Quantities in Oscillations

Physical Quantity	Symbol	Dimensions	SI Unit	Remarks
Period	T	$[\mathrm{T}]$	s	The least time for motion to repeat itself
Frequency	ν (or f)	$[\mathrm{T}^{-1}]$	s^{-1} (or Hz)	$\nu = \frac{1}{T}$
Angular frequency	ω	$[\mathrm{T}^{-1}]$	$\mathrm{rad\,s}^{-1}$ (or s^{-1})	$\omega = 2\pi\nu = \frac{2\pi}{T}$
Phase constant	ϕ	Dimensionless ($[\mathrm{M}^0\mathrm{L}^0\mathrm{T}^0]$)	rad	Initial value of phase of displacement in SHM at $t = 0$
Force constant	k	$[\mathrm{MT}^{-2}]$	$\mathrm{N\,m}^{-1}$	Simple harmonic motion restoring force $F = -kx$

10. NEET UG Practice Questions with Complete Worked Solutions

Note: The following original practice questions are strictly aligned with the NEET UG Physics syllabus and examination format to test conceptual depth, graphical analysis, and numerical problem-solving speed.

Practice Question 1 (Conceptual Classification)

Consider the following four physical motions:

I. A swimmer completing one return trip from one bank of a river to the other and back.

II. A freely suspended bar magnet displaced slightly from its North-South equilibrium direction and released.

III. A hydrogen molecule rotating uniformly about its centre of mass.

IV. The rotation of the Earth about its polar axis.

Which of the above motions are **periodic**, and which among them is **simple harmonic (for small displacements)**?

(A) Periodic: I, II, III, IV; Simple Harmonic: II only

(B) Periodic: II, III, IV only; Simple Harmonic: II only

(C) Periodic: II and III only; Simple Harmonic: II and III

(D) Periodic: II, III, IV only; Simple Harmonic: II and IV

Correct Answer: (B)

Worked Solution:

- **Statement I:** A swimmer completing a single return trip across a river does not repeat the motion indefinitely at regular intervals; hence it is *non-periodic*. (Similarly, an arrow released from a bow is non-periodic).
- **Statement II:** A freely suspended bar magnet displaced from its N-S magnetic meridian experiences a restoring magnetic torque $\tau = -MB\sin\theta \approx -MB\theta$ for small angles, so it executes *periodic and simple harmonic motion*.
- **Statement III & IV:** A hydrogen molecule rotating about its centre of mass and the Earth rotating about its axis repeat their motions at regular time intervals (T), so both are *periodic*, but neither moves to-and-fro about a mean equilibrium point, so they are *not simple harmonic*.

Practice Question 2 (Mathematical Identification of SHM)

Which of the following functions of time (ω is a positive constant) represents a **Simple Harmonic Motion**, and what is its time period?

(A) $x(t) = 3\cos\left(\frac{\pi}{4} - 2\omega t\right)$, with period $T = \frac{\pi}{\omega}$

(B) $x(t) = \sin^3(\omega t)$, with period $T = \frac{2\pi}{\omega}$

(C) $x(t) = \cos\omega t + \cos 3\omega t + \cos 5\omega t$, with period $T = \frac{2\pi}{\omega}$

(D) $x(t) = \exp(-\omega^2 t^2)$, with period $T = \frac{\pi}{\omega}$

Correct Answer: (A)

Worked Solution:

- **Option (A):** Since $\cos(-\theta) = \cos\theta$, we can rewrite $x(t) = 3\cos\left(\frac{\pi}{4} - 2\omega t\right) = 3\cos\left(2\omega t - \frac{\pi}{4}\right)$. This is of the standard SHM form $A\cos(\omega' t + \phi)$ with angular frequency $\omega' = 2\omega$. Its period is $T = \frac{2\pi}{\omega'} = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$.
- **Option (B):** Using $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, we have $\sin^3\omega t = \frac{3}{4}\sin\omega t - \frac{1}{4}\sin 3\omega t$. Being a superposition of two different frequencies (ω and 3ω), it is periodic with period $\frac{2\pi}{\omega}$, but *not* simple harmonic.
- **Option (C):** Superposition of three different frequencies (ω, 3ω, 5ω); periodic with period $\frac{2\pi}{\omega}$, but *not* SHM.
- **Option (D):** $\exp(-\omega^2 t^2)$ decreases monotonically to zero as $t \to \infty$; it is *non-periodic*.

Practice Question 3 (Signs of Kinematic and Dynamic Quantities in SHM)

A particle is executing linear simple harmonic motion between two extreme points P and Q separated by $10\,\text{cm}$. Take the direction from P to Q as the positive x-direction and the midpoint O as the origin ($x = 0$). What are the signs of **velocity (v)**, **acceleration (a)**, and **restoring force (F)** on the particle when it is at a distance of $4\,\text{cm}$ from Q and moving towards P?

(A) $v > 0,\ a > 0,\ F > 0$

(B) $v < 0,\ a < 0,\ F < 0$

(C) $v < 0,\ a > 0,\ F > 0$

(D) $v > 0,\ a < 0,\ F < 0$

Correct Answer: (B)

Worked Solution:
Since the total distance between extremes P and Q is $10\,\text{cm}$, the midpoint O is at $5\,\text{cm}$ from either end. Taking $P \to Q$ as positive, P is at $x = -5\,\text{cm}$, O is at $x = 0$, and Q is at $x = +5\,\text{cm}$.

- When the particle is $4\,\text{cm}$ away from Q, its position coordinate is $x = +5\,\text{cm} - 4\,\text{cm} = +1\,\text{cm} > 0$ (it lies between O and Q).
- Since it is moving towards P (in the direction opposite to $P \to Q$), its **velocity v is negative ($v < 0$)**.
- Since $x = +1\,\text{cm} > 0$ and $a = -\omega^2 x$ and $F = -kx$, both acceleration and restoring force are directed towards O (towards P), so $a < 0$ **and** $F < 0$.

Practice Question 4 (Initial Conditions & Phase Constant)

The motion of a particle executing SHM is described by $x(t) = A\cos(\omega t + \phi)$. At $t = 0$, the initial position of the particle is $x(0) = 1\,\text{cm}$ and its initial velocity is $v(0) = \omega\,\text{cm}\,\text{s}^{-1}$, where $\omega = \pi\,\text{s}^{-1}$. What are the amplitude A and initial phase angle ϕ of the particle?

(A) $A = \sqrt{2}\,\text{cm},\ \phi = +\frac{\pi}{4}\,\text{rad}$

(B) $A = \sqrt{2}\,\text{cm},\ \phi = -\frac{\pi}{4}\,\text{rad}$ (or $\frac{7\pi}{4}\,\text{rad}$)

(C) $A = 2\,\text{cm},\ \phi = -\frac{\pi}{3}\,\text{rad}$

(D) $A = 1\,\text{cm},\ \phi = 0\,\text{rad}$

Correct Answer: (B)

Worked Solution:

Given $x(t) = A\cos(\omega t + \phi)$, differentiating gives $v(t) = -\omega A\sin(\omega t + \phi)$. Applying the initial conditions at $t = 0$:

$$x(0) = A\cos\phi = 1\,\text{cm} \quad \text{--- (1)}$$

$$v(0) = -\omega A\sin\phi = \omega\,\text{cm}\,\text{s}^{-1} \implies A\sin\phi = -1\,\text{cm} \quad \text{--- (2)}$$

Squaring and adding (1) and (2):

$$A^2(\cos^2\phi + \sin^2\phi) = 1^2 + (-1)^2 = 2 \implies A = \sqrt{2}\,\text{cm}$$

Dividing (2) by (1):

$$\tan\phi = -1 \quad \text{with } \cos\phi > 0,\ \sin\phi < 0\ (\text{fourth quadrant}) \implies \phi = -\frac{\pi}{4}\,\text{rad}\ \left(\text{or } \frac{7\pi}{4}\,\text{rad}\right)$$

(Note: If written in sine form $x(t) = B\sin(\omega t + \alpha)$, then $B\sin\alpha = 1$ and $B\cos\alpha = 1$, giving $B = \sqrt{2}\,\text{cm}$ and $\alpha = +\frac{\pi}{4}\,\text{rad}$.)

Practice Question 5 (Spring Balance Scale & Oscillations)

A vertical spring balance has a linear scale that reads from 0 to $50\,\text{kg}$, and the length of the scale is $20\,\text{cm}$. A body suspended from this balance, when displaced slightly vertically and released, oscillates with a period of $0.6\,\text{s}$. What is the weight of the body? (Take $g = 9.8\,\text{m}\,\text{s}^{-2}$)

(A) $98.0\,\text{N}$

(B) $145.2\,\text{N}$

(C) $219.1\,\text{N}$

(D) $490.0\,\text{N}$

Correct Answer: (C)

Worked Solution:

A reading of $M_{\text{max}} = 50\,\text{kg}$ corresponds to a maximum stretching force $F_{\text{max}} = M_{\text{max}}g = 50 \times 9.8 = 490\,\text{N}$ over a maximum extension $x_{\text{max}} = 20\,\text{cm} = 0.20\,\text{m}$. Therefore, the spring constant k of the spring balance is:

$$k = \frac{F_{\text{max}}}{x_{\text{max}}} = \frac{490\,\text{N}}{0.20\,\text{m}} = 2450\,\text{N}\,\text{m}^{-1}$$

Using the time period relation $T = 2\pi\sqrt{\frac{m}{k}}$ for the oscillating body of mass m:

$$m = \frac{kT^2}{4\pi^2} = \frac{2450 \times (0.6)^2}{4 \times (3.1416)^2} = \frac{882}{39.478} \approx 22.34\,\text{kg}$$

Hence, the weight of the body is:

$$W = mg = 22.34\,\text{kg} \times 9.8\,\text{m}\,\text{s}^{-2} \approx 219\,\text{N}$$

Practice Question 6 (Horizontal Spring-Mass Oscillator Kinematics)

A spring having a spring constant $k = 1200\,\mathrm{N\,m^{-1}}$ is mounted on a horizontal frictionless table. A mass of $3\,\mathrm{kg}$ is attached to the free end of the spring, pulled sideways to a distance of $2.0\,\mathrm{cm}$, and released from rest. What are (i) the frequency of oscillations, (ii) the maximum acceleration of the mass, and (iii) the maximum speed of the mass?

(A) $\nu = 3.18\,\mathrm{Hz},\ a_{\max} = 8.0\,\mathrm{m\,s^{-2}},\ v_{\max} = 0.40\,\mathrm{m\,s^{-1}}$

(B) $\nu = 6.36\,\mathrm{Hz},\ a_{\max} = 16.0\,\mathrm{m\,s^{-2}},\ v_{\max} = 0.80\,\mathrm{m\,s^{-1}}$

(C) $\nu = 3.18\,\mathrm{Hz},\ a_{\max} = 0.40\,\mathrm{m\,s^{-2}},\ v_{\max} = 8.0\,\mathrm{m\,s^{-1}}$

(D) $\nu = 20.0\,\mathrm{Hz},\ a_{\max} = 8.0\,\mathrm{m\,s^{-2}},\ v_{\max} = 0.40\,\mathrm{m\,s^{-1}}$

Correct Answer: (A)

Worked Solution:

Given: $k = 1200\,\mathrm{N\,m^{-1}}$, $m = 3\,\mathrm{kg}$, and amplitude $A = 2.0\,\mathrm{cm} = 0.02\,\mathrm{m}$.

First, find the angular frequency ω:

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{1200}{3}} = \sqrt{400} = 20\,\mathrm{rad\,s^{-1}}$$

- **(i) Frequency of oscillations (ν):**

$$\nu = \frac{\omega}{2\pi} = \frac{20}{2 \times 3.1416} \approx 3.18\,\mathrm{Hz}$$

- **(ii) Maximum acceleration ($a_{\max}$):**

$$a_{\max} = \omega^2 A = (20\,\mathrm{s^{-1}})^2 \times (0.02\,\mathrm{m}) = 400 \times 0.02 = 8.0\,\mathrm{m\,s^{-2}}$$

- **(iii) Maximum speed ($v_{\max}$):**

$$v_{\max} = \omega A = (20\,\mathrm{s^{-1}}) \times (0.02\,\mathrm{m}) = 0.40\,\mathrm{m\,s^{-1}}$$

Practice Question 7 (Writing Displacement Functions for Different Starting Positions)

For the oscillator in Practice Question 6 ($\omega = 20\,\text{rad}\,\text{s}^{-1}$, $A = 2.0\,\text{cm}$, taking left-to-right as the $+x$-direction and unstretched position as $x = 0$), match the starting condition at $t = 0$ with the correct displacement function $x(t)$ (in cm):

- **Case I:** At $t = 0$, the mass is at the mean position moving towards the right ($+x$).
- **Case II:** At $t = 0$, the mass is released from rest at the maximum stretched position ($x = +2.0\,\text{cm}$).
- **Case III:** At $t = 0$, the mass is released from rest at the maximum compressed position ($x = -2.0\,\text{cm}$).

(A) I: $x = 2\sin 20t$; II: $x = 2\cos 20t$; III: $x = -2\cos 20t$

(B) I: $x = 2\cos 20t$; II: $x = 2\sin 20t$; III: $x = -2\sin 20t$

(C) I: $x = -2\sin 20t$; II: $x = 2\cos 20t$; III: $x = 2\sin 20t$

(D) I: $x = 2\sin 20t$; II: $x = -2\cos 20t$; III: $x = 2\cos 20t$

Correct Answer: (A)

Worked Solution:
All three motions have the exact same amplitude ($A = 2\,\text{cm}$) and angular frequency ($\omega = 20\,\text{rad}\,\text{s}^{-1}$); they differ *only in their initial phase* ϕ:

- **Case I ($x(0) = 0, v(0) > 0$):** Using $x(t) = A\cos(\omega t + \phi)$ with $\phi = -\frac{\pi}{2}$ gives $x(t) = 2\cos\left(20t - \frac{\pi}{2}\right) = 2\sin 20t\,\text{cm}$.
- **Case II ($x(0) = +2\,\text{cm}, v(0) = 0$):** Here $\phi = 0$, so $x(t) = 2\cos 20t\,\text{cm}$.
- **Case III ($x(0) = -2\,\text{cm}, v(0) = 0$):** Here $\phi = \pi$, so $x(t) = 2\cos(20t + \pi) = -2\cos 20t\,\text{cm}$.

Practice Question 8 (Reference Circle Parameters)

A simple harmonic motion is given by the equation:

$$x(t) = -2\sin\left(3t + \frac{\pi}{3}\right) \quad (\text{where } x \text{ is in cm and } t \text{ is in s})$$

If this SHM is represented as the x-projection of a reference particle rotating **anticlockwise** on a reference circle, what are the radius A of the circle, the angular speed ω, and the initial angle ϕ made by the radius vector with the $+x$-axis at $t = 0$?

(A) $A = 2\,\text{cm},\ \omega = 3\,\text{rad}\,\text{s}^{-1},\ \phi = \frac{\pi}{3}\,\text{rad}$

(B) $A = 2\,\text{cm},\ \omega = 3\,\text{rad}\,\text{s}^{-1},\ \phi = \frac{5\pi}{6}\,\text{rad}$

(C) $A = 2\,\text{cm},\ \omega = 3\,\text{rad}\,\text{s}^{-1},\ \phi = -\frac{\pi}{6}\,\text{rad}$

(D) $A = 3\,\text{cm},\ \omega = 2\,\text{rad}\,\text{s}^{-1},\ \phi = \frac{5\pi}{6}\,\text{rad}$

Correct Answer: (B)

Worked Solution:

To match the standard x-projection form $x(t) = A\cos(\omega t + \phi)$ with $A > 0$, we use the identity $-\sin\theta = \cos\left(\theta + \frac{\pi}{2}\right)$:

$$x(t) = -2\sin\left(3t + \frac{\pi}{3}\right) = 2\cos\left(3t + \frac{\pi}{3} + \frac{\pi}{2}\right) = 2\cos\left(3t + \frac{5\pi}{6}\right)$$

Comparing with $x(t) = A\cos(\omega t + \phi)$, we obtain:

- Radius of reference circle $A = 2\,\text{cm}$
- Angular speed $\omega = 3\,\text{rad}\,\text{s}^{-1}$
- Initial phase angle $\phi = \frac{5\pi}{6}\,\text{rad}$ (150°, in the second quadrant).

Practice Question 9 (Locomotive Piston Stroke and Maximum Speed)

The piston in the cylinder head of a locomotive has a stroke (which is defined as twice the amplitude) of $1.0\,\mathrm{m}$. If the piston moves with simple harmonic motion at an angular frequency of $200\,\mathrm{rad\,min^{-1}}$, what is its maximum speed?

(A) $3.33\,\mathrm{m\,s^{-1}}$

(B) $1.67\,\mathrm{m\,s^{-1}}$

(C) $100\,\mathrm{m\,s^{-1}}$

(D) $0.83\,\mathrm{m\,s^{-1}}$

Correct Answer: (B)

Worked Solution:

Given stroke $= 2A = 1.0\,\mathrm{m} \implies$ amplitude $A = 0.5\,\mathrm{m}$.

Convert angular frequency ω from $\mathrm{rad\,min^{-1}}$ to $\mathrm{rad\,s^{-1}}$:

$$\omega = 200\,\mathrm{rad\,min^{-1}} = \frac{200}{60}\,\mathrm{rad\,s^{-1}} = \frac{10}{3}\,\mathrm{rad\,s^{-1}}$$

The maximum speed of the piston is:

$$v_{\max} = \omega A = \left(\frac{10}{3}\,\mathrm{rad\,s^{-1}}\right) \times (0.5\,\mathrm{m}) = \frac{5}{3}\,\mathrm{m\,s^{-1}} \approx 1.67\,\mathrm{m\,s^{-1}}$$

Practice Question 10 (Simple Pendulum on the Moon)

The acceleration due to gravity on the surface of the Moon is $1.7\,\mathrm{m\,s^{-2}}$. What is the time period of a simple pendulum on the surface of the Moon if its time period on the surface of the Earth is $3.5\,\mathrm{s}$? (Take g on the surface of the Earth as $9.8\,\mathrm{m\,s^{-2}}$)

(A) $8.4\,\mathrm{s}$

(B) $1.46\,\mathrm{s}$

(C) $20.2\,\mathrm{s}$

(D) $6.1\,\mathrm{s}$

Correct Answer: (A)

Worked Solution:

The time period of a simple pendulum of fixed length L is $T = 2\pi\sqrt{\frac{L}{g}}$, which implies $T \propto \frac{1}{\sqrt{g}}$. Therefore, comparing the Moon (T_M, g_M) and the Earth (T_E, g_E):

$$\frac{T_M}{T_E} = \sqrt{\frac{g_E}{g_M}}$$

Substituting $T_E = 3.5\,\mathrm{s}$, $g_E = 9.8\,\mathrm{m\,s^{-2}}$, and $g_M = 1.7\,\mathrm{m\,s^{-2}}$:

$$T_M = 3.5\,\mathrm{s} \times \sqrt{\frac{9.8}{1.7}} = 3.5 \times \sqrt{5.7647} \approx 3.5 \times 2.401 \approx 8.4\,\mathrm{s}$$

Practice Question 11 (Assertion–Reason on Energy and Superposition in SHM)

Assertion (A): In a linear simple harmonic oscillator with time period T, both kinetic energy and potential energy are periodic functions of time with time period $T/2$.
Reason (R): During each complete oscillation of period T, the particle passes through the mean position twice and reaches the extreme positions ($\pm A$) twice, and both kinetic and potential energies depend on the squares of velocity and displacement respectively.

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Correct Answer: (A)

Worked Solution:
Since $K = \frac{1}{2}mv^2 = \frac{1}{2}kA^2\sin^2(\omega t + \phi) = \frac{1}{4}kA^2[1 - \cos(2\omega t + 2\phi)]$ and $U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \phi) = \frac{1}{4}kA^2[1 + \cos(2\omega t + 2\phi)]$, the sign of v and x is squared away. Both K and U peak twice per period T (at angular frequency 2ω), so their period is $\frac{2\pi}{2\omega} = \frac{T}{2}$. Thus both A and R are true and R correctly explains A.

Practice Question 12 (Displacement Where Kinetic and Potential Energies Are Equal)

A particle executes simple harmonic motion with amplitude A and time period T. Starting from the mean position $x = 0$ at $t = 0$, what is the **shortest time** taken by the particle to reach a point where its kinetic energy is equal to its potential energy?

(A) $\frac{T}{4}$

(B) $\frac{T}{6}$

(C) $\frac{T}{8}$

(D) $\frac{T}{12}$

Correct Answer: (C)

Worked Solution:
Setting kinetic energy equal to potential energy ($K = U$):

$$\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2 \implies 2x^2 = A^2 \implies x = \pm\frac{A}{\sqrt{2}}$$

Since the particle starts at $x = 0$ at $t = 0$ moving towards $+A$, its displacement equation is $x(t) = A\sin\omega t$. Setting $x(t) = \frac{A}{\sqrt{2}}$:

$$A\sin\left(\frac{2\pi}{T}t\right) = \frac{A}{\sqrt{2}} \implies \frac{2\pi}{T}t = \frac{\pi}{4} \implies t = \frac{T}{8}$$

11. Common Pitfalls & Quick Revision Checklist

Common Exam Pitfalls to Avoid in NEET UG:

- **Confusing Periodic Motion with SHM:** Remember that every SHM is periodic, and every oscillatory motion is periodic, but *not* every periodic motion is oscillatory or simple harmonic. Only motion governed by the linear restoring force $F = -kx$ ($a = -\omega^2 x$) is SHM. Relationships like $a = +0.7x$, $a = -200x^2$, or $a = 100x^3$ do *not* produce SHM; only $a = -10x$ (negative and first power of x) represents SHM.
- **Amplitude Independence of Time Period:** In a linear harmonic oscillator, the time period $T = 2\pi\sqrt{m/k}$ does *not* depend on the amplitude A, total energy E, or initial phase ϕ. (Contrast this with planetary orbits under gravity where period depends on orbit size by Kepler's third law, or a simple pendulum at large angles where $\sin\theta \not\approx \theta$).
- **Frequency of Energy vs. Frequency of Displacement:** If displacement $x(t)$ oscillates with frequency ν and period T, both kinetic energy $K(t)$ and potential energy $U(t)$ oscillate with frequency 2ν and period $T/2$, whereas total mechanical energy $E = K + U$ has zero frequency (remains constant).
- **Seconds Pendulum Period:** A "seconds pendulum" ticks once per half-oscillation ($1\,\text{s}$ from left extreme to right extreme), so its full time period is $T = 2\,\text{s}$ (and length $L \approx 1\,\text{m}$ on Earth), *not* $1\,\text{s}$.
- **Combination of Two SHMs:** A superposition of two SHMs with arbitrary frequencies is *not* necessarily periodic; it is periodic only if the ratio of their frequencies is a rational number (or one frequency is an integral multiple of the fundamental frequency).

Concise Final Recap

1. **Period & Frequency:** $\nu = \frac{1}{T}, \omega = \frac{2\pi}{T} = 2\pi\nu$.
2. **Displacement in SHM:** $x(t) = A\cos(\omega t + \phi)$ or $x(t) = A\cos\omega t + B\sin\omega t$.
3. **Velocity in SHM:** $v(t) = -\omega A\sin(\omega t + \phi) = \pm\omega\sqrt{A^2 - x^2}$; $v_{\max} = \omega A$ at $x = 0$. Phase leads $x(t)$ by $\frac{\pi}{2}$.
4. **Acceleration in SHM:** $a(t) = -\omega^2 A\cos(\omega t + \phi) = -\omega^2 x(t)$; $a_{\max} = \omega^2 A$ at $x = \pm A$. Phase leads $x(t)$ by π.
5. **Restoring Force & Linear Oscillator:** $F(t) = -kx(t)$ where $k = m\omega^2 \implies T = 2\pi\sqrt{\frac{m}{k}}$.
6. **Mechanical Energy in SHM:** $K = \frac{1}{2}k(A^2 - x^2)$, $U = \frac{1}{2}kx^2$, and Total Energy $E = K + U = \frac{1}{2}kA^2 = \text{constant}$.
7. **Simple Pendulum (Small Angles $\theta \lesssim 15^\circ$):** $\alpha = -\frac{mgL}{I}\theta \implies T = 2\pi\sqrt{\frac{I}{mgL}} = 2\pi\sqrt{\frac{L}{g}}$.